Vivek Gupta

Backend Engineer · API Development · Distributed Systems · RAG & LLM Systems

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PROFESSIONAL SUMMARY

Backend Engineer with **4 years** designing and building distributed systems, event-driven microservices, REST APIs, and workflow automation in enterprise production environments. Sole backend engineer at PPI Advisory, a national Canadian financial advisory firm, owning system design, testing, CI/CD, and production support for services used by **600+ employees**. Proficient in C#, .NET, Node.js, TypeScript, Python, SQL Server, MongoDB, and Azure. Most recently, solo-built and deployed a production RAG system (Python, FastAPI, ChromaDB, Claude API) with a hand-built evaluation harness scoring **94% generation accuracy** and **zero hallucination**.

KEY ACHIEVEMENTS

- Sole engineer who designed, architected, and deployed a distributed, event-driven email classification system — processing **60,000–80,000 emails/day** across **600 employees** using Microsoft Graph API webhooks, with exponential backoff retry logic for fault tolerance, eliminating daily manual triaging company-wide.
- Built and deployed a retrieval-augmented generation (RAG) system from scratch that answers questions over FastAPI's official documentation, achieving **79% retrieval accuracy** and **94% generation accuracy** on an 18-question evaluation set with **zero hallucinations** on out-of-scope queries.
- Delivered a high-reliability backend service end-to-end (stakeholder alignment, system design, unit testing, code review, CI/CD deployment, Azure monitoring) — replacing a costly third-party plugin with a scalable in-house microservice at **a fraction of the cost**.
- Designed and owned a concurrent data processing system (SQL Server commission engine) with structured error handling, execution logging, query result caching, and performance-optimised stored procedures — supporting a nationwide network of **hundreds of independent financial advisors**.

PROJECTS

RAG Docs Assistant · Python · FastAPI · ChromaDB · Anthropic Claude API · Docker · 2026

- Live Demo ↗ (free-tier host — may take ~30-60s to wake up on first load) GitHub ↗
- Built and deployed a retrieval-augmented generation (RAG) system from scratch that answers questions over FastAPI's official documentation, achieving **79% retrieval accuracy** and **94% generation accuracy** on an 18-question evaluation set with **zero hallucinations** on out-of-scope queries.
- Designed and implemented the full pipeline: document ingestion and chunking (**155 files, ~1,900 chunks**), vector embeddings and storage with ChromaDB, metadata-based retrieval filtering, and grounded generation via the Anthropic Claude API with **source citation**.
- Containerized the application with Docker and deployed it publicly on Render with per-IP rate limiting to control API costs.
- Diagnosed and resolved **four separate production issues** — a Python version dependency conflict, a Docker image caching gap, a data-store naming collision causing a **live outage**, and a reverse-proxy IP-detection bug breaking rate limiting — each root-caused and fixed solo.
- Built a live interactive frontend with a floating chat widget, real-time markdown-rendered answers, and example queries demonstrating grounded refusal behavior.

SaleSpotter · Node.js · Express.js · React · TypeScript · MongoDB Atlas · JWT · REST API

- Designed and built a **full-stack marketplace platform** — architected scalable REST APIs with clearly defined HTTP endpoints and request/response contracts for authentication, ad management, notifications, and geospatial search, and built the React frontend consuming them. Implemented **JWT-based authentication and role-based access control**, plus MongoDB geospatial queries for location-based discovery.

PROFESSIONAL EXPERIENCE

Backend Engineer **Feb 2022 – Dec 2025**

PPI Advisory · Toronto, ON, Canada · High-net-worth division of PPI, a national Canadian insurance marketing organization

Distributed Email Classification Service · C# · .NET · Microsoft Graph API · SQL Server · Azure

- Identified a critical performance bottleneck in the existing Outlook plugin — **independently proposed, architected, and delivered** a distributed event-driven microservice approved by stakeholders after a formal system design and architecture review.
- Designed a concurrent, webhook-based data processing pipeline over HTTP: Microsoft Graph API subscriptions delivered email event notifications to the backend's REST endpoints, triggering the full request/response lifecycle — sender lookup against SQL Server, revenue-based priority scoring, and classification tag application via the Graph API — **entirely server-side with no local compute**.
- Implemented exponential backoff retry logic and fault tolerance mechanisms to handle Microsoft Graph API rate limits — automatically retrying at increasing intervals up to 2 hours before escalating to an error state, ensuring **zero email classification loss** during API throttling windows.
- Processed **60,000–80,000 emails/day** across **600 employees** with consistent reliability; system operated uniformly across Outlook desktop, web, and mobile, extending support to shared mailboxes unavailable in the previous architecture.
- Monitored system health using Azure Monitor and application log analysis — proactively identifying anomalies, diagnosing root causes, and maintaining high availability of the production service.

Commission Calculation Engine & Financial Reporting API · SQL Server · C# · .NET · Power BI

- Architected and maintained a concurrent data processing system of SQL Server stored procedures for daily advisor commission calculations — incorporating performance-optimised queries, temp table caching, structured error handling, and execution status logging; **validated stored procedure logic with unit tests before each production release**.
- Applied scalability and reliability principles to handle multi-company rate structures, secretary email attribution, and year-end tier-based payout classifications for **hundreds of financial advisors across a national network**.

- Built and deployed a backend REST API with well-documented endpoints consumed by the finance team's internal web portal — delivering **real-time per-advisor commission data** and reducing dependency on manual spreadsheet reporting.

Backend Automation & Data Processing Systems · PowerShell · Azure · Active Directory · SQL Server

- Designed and built a fault-tolerant, resumable data archival system that crawled **~200 GB of hybrid on-prem/Azure storage** — identifying files by access age, migrating them with full directory structure preservation, and maintaining a complete audit log; **built to resume from checkpoint if interrupted mid-run**.
- Developed a distributed identity sync system linking **three enterprise applications** on a shared key — handling concurrent updates, post-marriage email change edge cases, auto-pausing **100–150 subscriptions** for employees on leave, and generating HR reconciliation reports.
- Built a service account lifecycle management system integrated with Active Directory — automatically disabling inactive accounts and **reducing licensing overhead** across the organisation.
- Developed and maintained **50+ PowerShell backend scripts** across finance, HR, and operations — all built with OOP design principles, structured logging, and fault-tolerant error handling; participated in code reviews to ensure script reliability before deployment.

Enterprise Application Support & Observability

- Monitored production backend services using Azure Monitor and IIS application log analysis — proactively identifying performance degradation and **resolving incidents before business impact**.
- Managed Agile delivery using Jira — tracking sprint tasks, reporting progress in weekly stakeholder meetings, and maintaining clear visibility of backend development status across the full SDLC.

CERTIFICATIONS

Microsoft Certified: Azure Developer Associate (AZ-204) — In Progress

Anthropic Academy — AI Fluency: AI Capabilities & Limitations, Claude Platform 101, Claude with the Anthropic API, Introduction to Model Context Protocol (MCP), Claude Code 101

CORE SKILLS

Languages: C#, JavaScript, TypeScript, Python, SQL, PowerShell

AI / LLM: RAG System Design, Vector Search & Embeddings (ChromaDB), LLM Evaluation Methodology, Prompt Engineering, Anthropic Claude API

Backend: .NET/C#, Node.js, Express.js, FastAPI, REST API Design, HTTP & Request/Response Lifecycle, Microservices, Event-Driven Architecture, Webhooks, Microsoft Graph API, JWT Authentication & Authorization, Retry Logic & Fault Tolerance

Data Processing: SQL Server (stored procedures, query optimisation, temp table caching, concurrent processing), MongoDB Atlas, ChromaDB, Relational Databases, Data Pipelines

Cloud & DevOps: Microsoft Azure, Azure Monitor, Docker, CI/CD Pipelines, Git, GitHub, Active Directory, IIS, Postman, Power BI, Render

Quality & Observability: Unit Testing, Code Reviews, Structured Logging, Application Log Analysis, Azure Monitor, Production Incident Response, Performance Monitoring

Concepts: System Design, Object-Oriented Programming, Scalability Principles, Performance Optimisation, Reliability Engineering, Fault Tolerance, Concurrency, Backend Architecture, Agile / Jira, Full SDLC, Identity & Access Management

EDUCATION

Honours Bachelor of Computer Science — Sheridan College, Oakville, ON, Canada · Dec 2021 · GPA 3.00/4.00